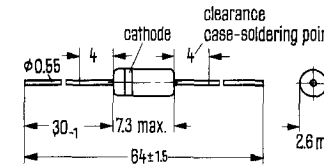
General-purpose silicon diodes

Silicon diodes BAY 44, BAY 45 and BAY 46 are suitable for universal application in equipment with high operating temperatures and where space is at a premium. They are provided with a glass-case 51 A 2 DIN 41880 (DO-7). The cathode end is marked by a colour ring.

Type	Order number
BAY 44	Q 60201-Y 44
BAY 45	Q 60201-Y 45
BAY 46	Q 60201-Y 45

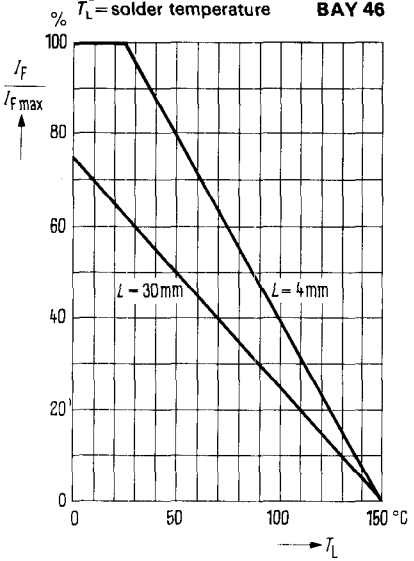


Weight approx. 0.2 g Dimensions in mm

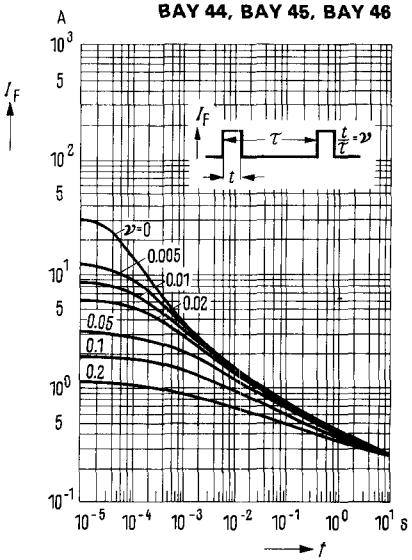
Maximum ratings (T_{amb} = 25 °C)		BAY 44	BAY 45	BAY 46	
Reverse voltage	V_R	50	150	300	V
Maximum reverse voltage	v_{RM}	50	150	300	V
Forward current (L = 4 mm; see diagram)	I_F		250		mA
Maximum forward current (t = 10 µs see diagram)	i_{FM}		30		A
Junction temperature	T_j		150		°C
Ambient temperature	T_{amb}		−55 to +125		°C
Total power dissipation (T_{amb} = 25 °C; L = 4 mm)	P_{tot}		250		mW
Thermal resistance (L = 4 mm)[2]	R_{thJamb}		<380		K/W
Static characteristics					
Forward voltage (I_F = 100 mA; T_{amb} = 25 °C)	V_F		0.97 (<1.1)*		V
Forward voltage (I_F = 100 mA; T_{amb} = 100 °C)	V_F		0.9		V
Reverse current ($V_R = V_{RM}$; T_{amb} = 25 °C)	I_R		0.02 (<0.2)*		µA
Reverse current ($V_R = V_{RM}$; T_{amb} = 100 °C)	I_R		0.4 (<10)		µA
Dynamic characteristics					
Capacitance (V_R = 0 V; f = 1 MHz)	C_O		7		pf
Capacitance (V_R = 5 V; f = 1 MHz)	C_s		2.5		pf
Reverse recovery time when switching from I_F = 5 mA to I_R = 2 mA[1])	t_{rr}		4.5		µs
Detector voltage efficiency (V_{eff} = 5 V; f = 1 MHz; R_L = 10 kΩ; C_L = 10 nf)	η_v		65		%

[1]) Measured with Tektronix type S plug-in unit * AQL = 0.65%
[2]) *These value apply to a case-soldering point clearance of 4 mm*

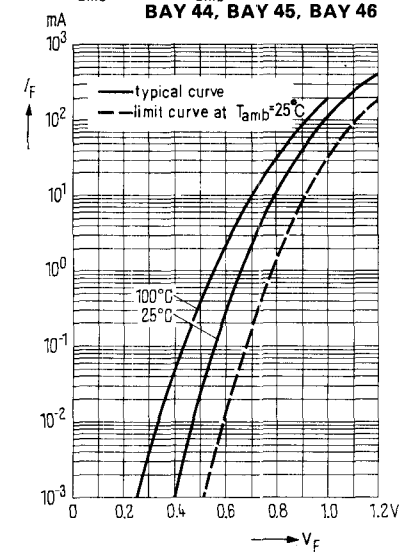
Max. permissible forward current
$I_F/I_{F\,max} = f(T_L)$ **BAY 44**
L_L = distance case to solder: **BAY 45**
T_L = solder temperature **BAY 46**



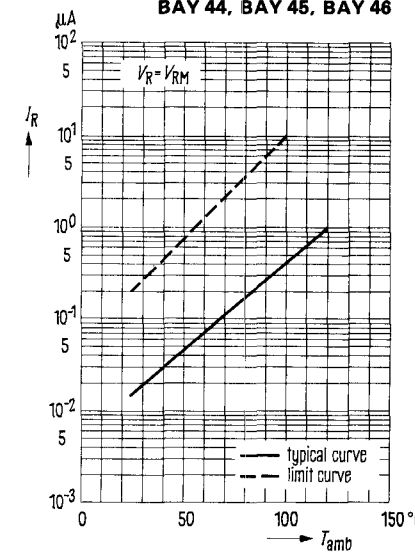
Permissible pulse load $I_F = f(t)$
ν = parameter; T_{amb} = 25 °C
BAY 44, BAY 45, BAY 46



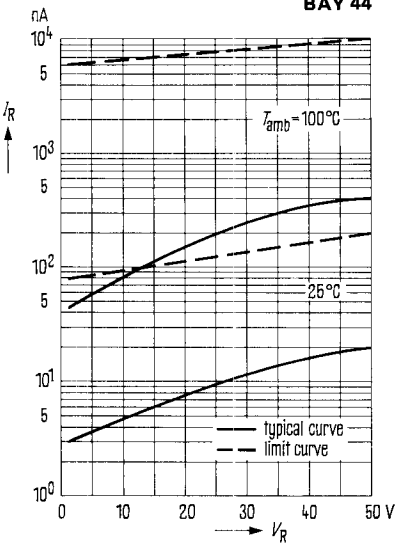
Forward characteristic $I_F = f(V_F)$
T_{amb} = 25 °C; T_{amb} = 100 °C
BAY 44, BAY 45, BAY 46



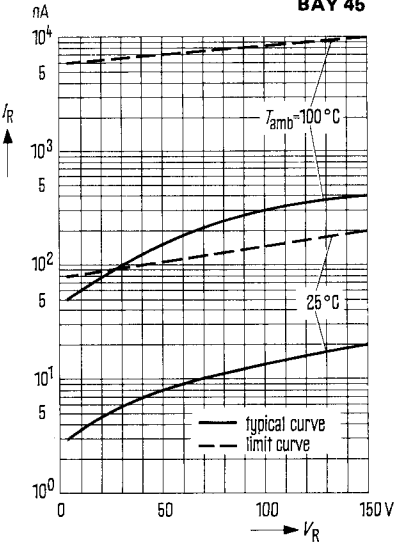
Reverse current $I_R = f(T_{amb})$
at max. permissible reverse voltage
BAY 44, BAY 45, BAY 46



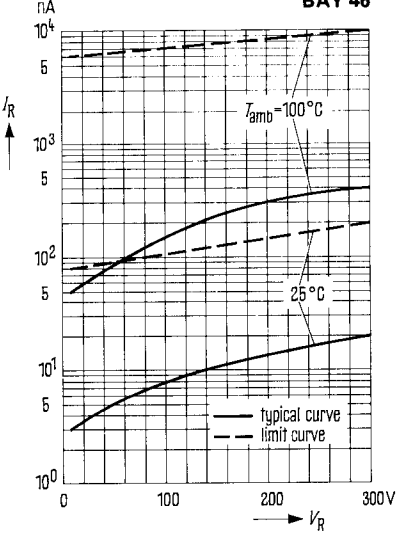
Reverse characteristics $I_R = f(V_R)$
$T_{amb} = 25\ °C$; $T_{amb} = 100\ °C$
BAY 44



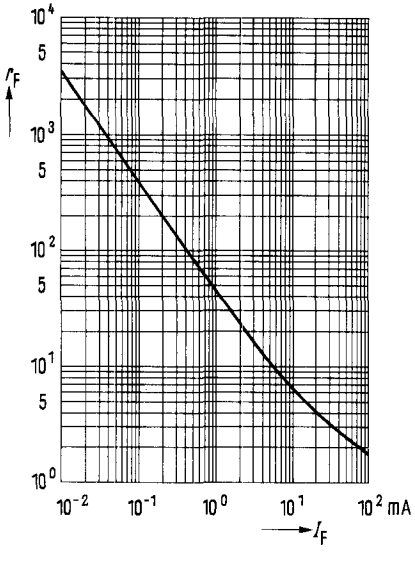
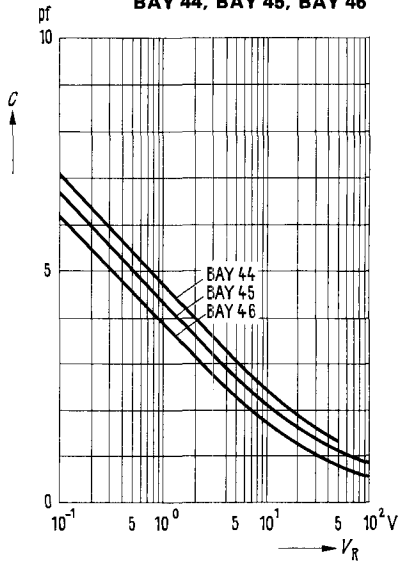
Reverse characteristics $I_R = f(V_R)$
$T_{amb} = 25\ °C$; $T_{amb} = 100\ °C$
BAY 45



Reverse characteristics $I_R = f(V_R)$
$T_{amb} = 25\ °C$; $T_{amb} = 100\ °C$
BAY 46



Dynamic forward resistance
$r_F = f(I_F)$; at $f = 1$ MHz
BAY 44, BAY 45, BAY 46



Junction capacitance as a function of reverse voltage
$C = f(V_R)$
BAY 44, BAY 45, BAY 46